# Corporate Finance

## A.Y.2025-2026

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# INTRODUCTION

## 1.1 Overview

Salesforce is a global leader in CRM software operating in a highly competitive and fast-evolving enterprise technology market. This sector is characterised by recurring revenues, high switching costs, and rapid innovation, factors that reward firms capable of combining high efficiency with an optimal capital structure. Since its foundation in 1999, Salesforce has expanded through continuous product development and strategic acquisitions, securing a dominant position in a market driven by growing investment in cloud infrastructure and AI-powered automation.

Until now, the company has maintained a low financial leverage, relying mainly on equity financing.

This is why we assume Salesforce as an underlevered firm for our analysis. After discussing alternative options with the professor, we adopted a target leverage ratio of 20%. Our project evaluates the implications of increasing Salesforce's leverage on this target. We begin by calibrating the long-term growth rate of free cash flows so that our valuation aligns with the current market price.

Using this baseline, we assess the type and pricing of securities needed to reach the new capital structure, the value created for existing shareholders, the post-transaction stock price and the firm's exposure to bankruptcy risk. Finally, we incorporate uncertainty through a Monte Carlo simulation to obtain the distribution of possible equity valuations.

## 1.2 Data sources and selection

In our analysis, we introduced several assumptions to ensure consistency and methodological rigor.

All assumptions were formulated after conducting detailed quantitative and qualitative assessments. Below, we outline each assumption and its underlying rationale.

- **Effective Tax Rate = 16.5%**: we estimated this effective tax rate based on the historical average. Although we observed outliers in past years, likely driven by deferred tax effects, we adopted a normalized, industry-consistent rate to avoid distortions and maintain realism in our forward-looking analysis.
- **Unlevered Cost of Capital (Ru) = 9,26%** : we computed the unlevered cost of capital using the standard CAPM formulation $Ru = Rf + \beta \times (Rm - Rf)$. Each input was derived through a structured and transparent process:
    - **Risk-Free Rate (Rf) = 4.07%**: Obtained from the 10-Year U.S. Government Bond Yield sourced from Bloomberg, reflecting a widely accepted proxy for long-term risk-free returns.
    - **Risk Premium = 4.25%**: We used the long-term historical average ERP for the U.S. market (1960–2024) from Professor Aswath Damodaran's dataset. A broad time window was intentionally selected to reduce cyclical bias and improve accuracy.
    - **Beta (β) = 1.22:** we performed an extensive beta analysis using multiple approaches to ensure robustness: We conducted a linear regression of monthly stock returns against the S&P 500 over the period 2020–2025. We computed beta using both raw returns and logarithmic returns; we selected the linear return regression as it aligns with typical analyst practice, while using the log-based estimate as a consistency check. Complemented this with a comparable companies beta analysis, following industry practice: first, identified representative firms in the sector using Capital IQ, than de-levered each comparable's beta to obtain asset betas and finally, computed the industry average asset beta. Moreover, we verified the published beta for Software (System & Application) from Professor Damodaran, finding that beta = 1.24. This finding supports the reliability of our estimate.

- **Fraction of Unlevered Firm Value Lost in Bankruptcy = 47.40%**: taken from the courses's guidelines, specifically the Appendix table "Fraction of Unlevered Firm Value Lost in Bankruptcy" selecting the category "Other" (see Appendix 1). This choice reflects the firm's characteristics more accurately than industry-specific extremes.
- **Recovery Rate in Case of Default = 41.0%**: We applied the same recovery rate used in class exercises to ensure methodological consistency with course conventions and analytical frameworks.

## 1.3 Determine the forecast and the financial statement

For building the Excel model, we incorporated historical data from the company's 10-K financial statements for the fiscal years 2018 to 2025. We forecasted the next 10 years (2025–2035). We selected a 10 year long projection period because Salesforce is a mature industry leader; a ten-year horizon allows us to model a gradual "soft landing" in revenue growth, transitioning smoothly from current double-digit rates to a stable long-term trajectory, thereby ensuring a more reliable terminal value calculation.

For our revenue growth projections, we combined market and industry analysis with expectations derived from S&P Capital IQ via the CEDIF database. We projected growth to decelerate from approximately 11.2% in the first forecast year down to 4.0% by 2035. . This approach was designed to align Salesforce's current high-growth trend with the stable long-term growth of the market, ensuring a smooth transition into the terminal value calculation without unrealistic discontinuities.

Regarding the cost structure, we assumed that key financial metrics would stabilize based on recent historical averages and industry benchmarks. We expect the Gross Profit Margin to remain robust at approximately 80.5%, consistent with the high-margin nature of cloud software subscriptions. Similarly, we modeled operating expenses, including R&D and marketing, to stabilize at roughly 61% of revenue. Operationally, this corresponds to applying relatively stable COGS and operating expense ratios to revenue across the forecast period, consistent with Salesforce's historical cost structure and the expectations embedded in analyst estimates.

For taxation, we applied an effective tax rate of approximately 16.5%, calculated as the average of the effective tax rates from previous comparable years, excluding outliers from the pandemic period to ensure a reasonable prediction.

## 1.4 Determining the long-term growth

We calculated the long-term growth rate to determine the terminal value of these cash flows. We defined Free Cash Flow as Cash Flow from Operations (CFO) minus Capital Expenditures. Specifically, we assumed the CFO would remain steady at 26% of revenue, predicted as the average of the past years.
The long-term growth rate of Free Cash Flows was calibrated so that the present value of the unlevered firm (Enterprise Value) aligns with the market's stock price valuation. To achieve this alignment, we added excess cash to the unlevered firm value ($V_U$), assuming excess cash grows at the risk-free rate. Excess cash is evaluated as total cash and cash equivalents minus the cash required for operations, estimated as 10% of total assets, consistent with standard corporate finance practice.
To obtain this rate, we computed the terminal value of Free Cash Flow as a growing perpetuity, based on the FCF projected for 2035:

$$TV_T = \frac{\overline{FCF}_T(1+g)}{r_U - g}$$



Unil.

We then used Excel's Goal Seek to find the value of g that makes the present value of the unlevered firm, plus excess cash, equal to Salesforce's market capitalization on our valuation date (10 October 2025). The target market capitalization was $247.91 billion. Through this iterative calibration, we obtained a long-term growth rate: $g = 3.43\%$ .

This rate is economically reasonable for two reasons. First, it is comfortably below the Risk-Free Rate (Rf= 4.07%), respecting the fundamental principle that a firm cannot grow faster than the economy in perpetuity. Second, it is consistent with a sustainable long-run growth rate roughly in line with expected GDP growth plus inflation.

## 1.5 Tax Shield Valuation

Following the calculation of the forecast free cash flows and the terminal value, our analysis focused on the impact of tax shields resulting from our hypothetical increase in the leverage ratio. We employed an iterative methodology to determine the optimal cost of debt and, subsequently, the associated interest payments, based on the firm's revised credit rating and corresponding credit spread.

Thanks to the continuous analysis of the updating coverage ratio, we were able to obtain the rating, the spread and the default probability for each year. In the initial iteration, the interest payment is calculated by multiplying the amount of debt issued by the risk-free rate to estimate a preliminary coverage ratio. In subsequent iterations, as we refined our assumptions, the interest payment was calculated by multiplying the outstanding debt balance by the yield on debt, computed by adding the credit spread to the prevailing risk-free rate.

Once interest payments were obtained, we calculated the annual interest tax shield as:

$$\text{Tax Shield}_t = \text{Interest}_t \times \text{Tax Rate}$$

Using an effective tax rate of approximately 16.5%, based on the average of Salesforce's historical effective tax rates (excluding outliers). To discount these tax shields under the Adjusted Present Value ( *APV* ) method with annually adjusted debt, we applied the Miles–Ezzell framework, which links the tax-shield discount rate $r_{TS,t}$ ) to the unlevered cost of capital ($r_U$) and the year-specific cost of debt ($r_{D,t}$). This ensures that the risk profile of the tax shields reflects the firm's rebalancing policy.

By discounting all annual tax shields and their terminal component with the appropriate discount rate, we obtained the present value of tax shields ( $PV_{TS}$ ), capturing the cumulative tax benefit of introducing and maintaining the 20% leverage ratio.

## 1.6 Cost of financial distress

To forecast the cost of financial distress, we used the Almeida-Philippon model (2008), which provides a robust framework for evaluating the costs associated with financial distress by combining credit spreads, recovery rates, and risk-neutral default probabilities. The model quantifies how market-implied debt yields and credit spreads reflect anticipated losses resulting from potential defaults. We computed the risk-neutral probability of default for each year using the standard relationship between spread, yield, and recovery rate (with a recovery rate of 41%, in line with empirical evidence and course guidance). We then determined the Present Value (PV) of distress costs recursively. We first calculated the distress costs in the terminal year (2035) as a steady state and then worked backward to the present using the recursive formula provided by the model.

$$\Phi = \frac{q\phi V_U + (1-q)\Phi}{1+r_f} \Rightarrow \Phi = \frac{q\phi V_U}{r_f + \phi}$$

The model assumes that, in the event of default, a fraction $\phi = 47.4\%$ of the unlevered firm value is lost. Working backward from the terminal year (steady state) to the present, we applied the recursive Almeida–Philippon formula, which combines: the expected loss in each year (default probability × $r_{D,t}$× unlevered firm value), and the probability-weighted continuation value of future distress costs, discounted at the cost of debt.

$$\Phi_t = \frac{q_t \phi V_{U,t} + (1 - q_t)\Phi_{t+1}}{1 + r_f}$$

## 1.7 Determine the leverage & the value of levered firm

Initially, we planned to evaluate a 40% leverage ratio, but preliminary simulations showed that such a high debt level would imply excessive default risk and unrealistic outcomes (including unstable costs of debt). After discussing with the professor, we revised our target to a more sustainable leverage ratio of 20%, which is still a significant increase compared to Salesforce's current very low leverage (around 1.8% of capital).

To move from the current low-debt structure to the target 20% leverage, we used Excel Solver to determine the optimal path of debt issuance. Solver was set to minimize the sum of squared deviations between the actual leverage ratio in each year and the 20% target.

This tool adjusted the debt level while simultaneously accounting for the interconnected financial impacts, specifically, how increased debt alters the credit rating, which changes the cost of debt, which in turn alters the tax shields and distress costs.

To maintain the target structure over time, we assumed annual rebalancing of debt: each year, debt is updated so that the leverage ratio remains close to 20% given the new value of the levered firm:

$$V_L = V_U + PVTS - PVBC$$

# REPORT QUESTIONS

## 2.1 What securities does the firm need to issue and at what price?

To implement the target capital structure of 20% debt, the firm will issue 10-year fixed-coupon corporate bonds with a face value of $1,000. We have chosen a strategy of annual issuances: in the first year (2025), a substantial volume of bonds will be issued to jump-start the leverage ratio from its current near-zero level to the 20% target. In subsequent years, the firm will issue smaller tranches of new bonds. These subsequent issuances serve to cover the increasing total debt required to maintain the constant 20% leverage ratio as the firm's value grows.

We selected a maturity of 10 years to align the debt profile with the long-term nature of Salesforce's assets and strategic horizon. As a major technology firm with significant investments in capitalized software, data centers, and long-term customer relationships (SaaS model), a decade-long maturity provides financial stability. It locks in the cost of capital for a substantial period, reducing the "refinancing risk" that would arise from relying on short-term commercial paper. Furthermore, selecting a fixed annual coupon structure offers predictability in cash outflows, allowing the firm to better forecast its interest coverage and manage liquidity.

The price of each bond is determined using the Discounted Cash Flow (DCF) method. The expected cash flows (annual coupons and final principal repayment) are discounted using the firm's estimated Cost of Debt for each respective year.

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It is important to note that while the coupon rate is fixed at the time of issuance (set equal to the cost of debt in the issuance year), the discount rate varies annually in our model as the firm's credit rating and spread evolve.

For the initial issuance in 2025:

- **Total Debt to Issue:** We plan to raise approximately $42,983.41 million to reach the 20% leverage target immediately.
- **Coupon Rate:** The bonds will carry a fixed annual coupon of 6.23%, which corresponds to the cost of debt estimated for Salesforce in 2025.
- **Bond Price:** The calculated issuance price is $1,020.38 per bond. This premium over the $1,000 face value reflects the market's expectation that Salesforce's credit risk (and thus its cost of debt) will likely decrease in future years (dropping to as low as 5.53% by 2029 ), making the fixed 6.23% coupon highly attractive.
- **Number of Securities:** To raise the required capital at this price, the firm will issue approximately 42.12 million bonds.

This pricing approach ensures that the capital raising reflects the true economic cost of borrowing and the market's view of Salesforce's creditworthiness over the life of the bond.

| Year | Bond Type | Coupon Rate | Bond Price | Debt to Issue (millions USD) | Number of Bonds (millions) |
|---|---|---|---|---|---|
| 2025 | Type 1 | 6.23% | 1,020.38 | 42,983.41 | 42.13 |
| 2026 | Type 2 | 6.69% | 1,061.79 | 3,724.42 | 3.51 |
| 2027 | Type 3 | 6.24% | 1,036.51 | 3,991.80 | 3.85 |
| 2028 | Type 4 | 6.32% | 1,048.67 | 4,602.17 | 4.39 |
| 2029 | Type 5 | 5.53% | 993.00 | 3,395.03 | 3.42 |
| 2030 | Type 6 | 6.32% | 1,055.61 | 2,537.59 | 2.40 |
| 2031 | Type 7 | 5.53% | 1,000.00 | 2,934.19 | 2.93 |
| 2032 | Type 8 | 5.53% | 1,000.00 | 2,204.26 | 2.20 |
| 2033 | Type 9 | 5.53% | 1,000.00 | 2,198.17 | 2.20 |
| 2034 | Type 10 | 5.53% | 1,000.00 | 4,049.94 | 4.05 |
| 2035 | Type 11 | 5.53% | 1,000.00 | 2,729.15 | 2.73 |
| **Total** | | | | **75,350.13** | **73.81** |

Tab 1: Bond issuance schedule

## 2.1 Will the transaction create value for existing shareholders?

No, the transaction does not generate value for current shareholders, at least not in the short term. Indeed, although Salesforce operates at a very steady and constant stream of cash flows due to its SaaS subscription model with high renewal rates and its dominant position, the stability of such cash flows is not enough to warrant more debt.

Yet, stable cash flows alone do not imply low financial distress costs or high collateral values.
Even moderate leverage drives credit spreads higher, cuts recovery rates, and raises the expected loss given default. In consequence, the incremental benefit of the interest tax shield is dominated by the probability of bankruptcy costs and the value of the levered firm falls short of the unlevered firm value for the near future. For the existing equity holders, this means a decrease in the total firm value and, therefore, a lower value of equity.
In the future, however, things change. Beyond 2030, forecasts suggest Salesforce's cash flows should increase, operating volatility decline and leverage become sustainable. It is under these circumstances that the interest tax shield starts to dominate the expected costs of financial distress and the value of the levered firm starts to exceed that of the unlevered firm. The implication of this is that leverage could start to create shareholder value in the long term, but this is delayed.
To conclude, the proposed transaction does not create value in the short term, but over the long term, as cash flows grow and financial risk declines, the tax shield could outweigh distress costs, allowing leverage to generate shareholder value.

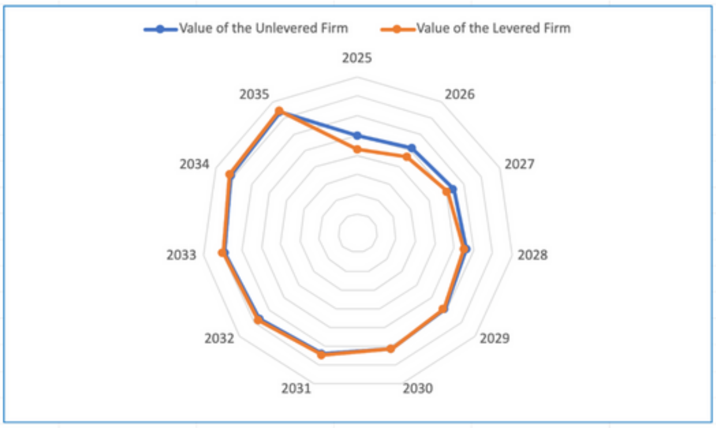


Figure 1: Comparison chart values levered and unlevered firms

## 2.3 What will be the value of stock at the end of all transactions? How does it compare with current stock price?

The resulting stock price after the transaction is $226.05, compared to the current price of $260.41.
To assess the impact of the debt-financed buyback on the firm's stock price, we first calculate the new stock price by dividing the value of the levered firm by the current number of shares, which results in a decline in price of approximately 13% per share. Next, we determine the number of shares that the company can repurchase using the newly issued debt. Dividing the debt issued by the new stock price, we find that the company can repurchase approximately 190 million shares (see Appendix 2).
Despite reducing the number of shares outstanding, the post-transaction stock price of $226.05 indicates that the negative effect of additional leverage outweighs the benefit of the buyback. In other words, the increase in debt reduces equity sufficiently that each share is worth less after the transaction.

## 2.4 How much will levering up expose the firm to bankruptcy risk?

Leveraging up to a 20% debt ratio exposes the firm to markedly higher bankruptcy risk in the short term, while producing a more balanced risk–benefit profile only in the long run.
Immediately after the recapitalization, the firm's credit rating drops to CCC, indicating very limited debt capacity and a materially elevated probability of default. This deterioration is reflected in the cost dynamics: from 2025 to 2029, the PV of bankruptcy costs significantly exceeds the PV of tax shields, with distress costs peaking at nearly $44 million in 2025 against roughly $11 million in tax benefits. These early years represent the most fragile phase for the firm, characterized by refinancing risk, restricted capital-market access and heightened uncertainty around solvency.
As operating performance strengthens and the firm progressively deleverages through growth, its risk profile improves substantially. From 2030 onward, expected distress costs decline below the value of the tax shields and by 2035 the company reaches an A- rating. In this more mature stage, the capital structure becomes increasingly sustainable: the tax benefits of debt are slightly larger than the remaining bankruptcy costs, indicating a much lower probability of financial distress.

However, while the long-term outlook becomes favorable, the short-term spike in financial risk dominates the net effect of the leverage decision. The heavy concentration of bankruptcy costs in the early years outweighs the later benefits, making the overall impact of the 20% leverage ratio value-reducing.

In summary, the firm ultimately stabilizes and benefits from the tax shield, but the initial period of severe credit deterioration renders the leverage choice inefficient when evaluated over the full horizon.

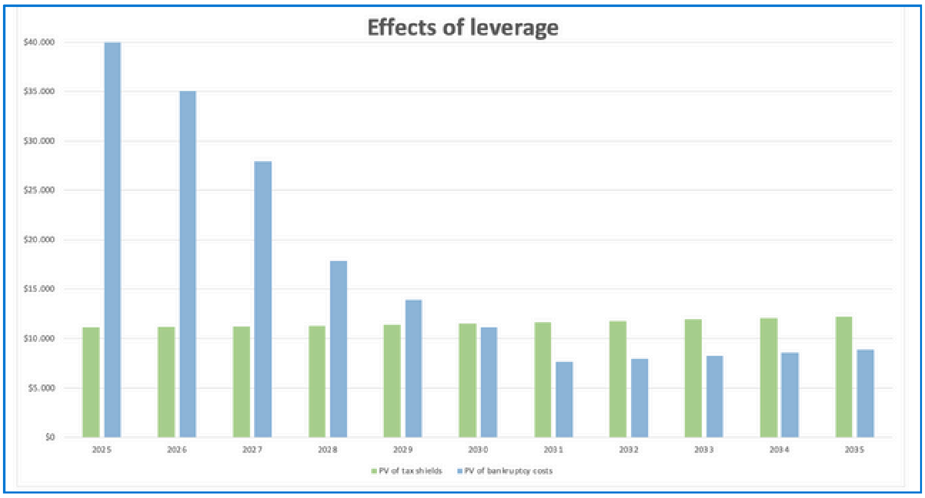


Figure 2: Effects of leverage


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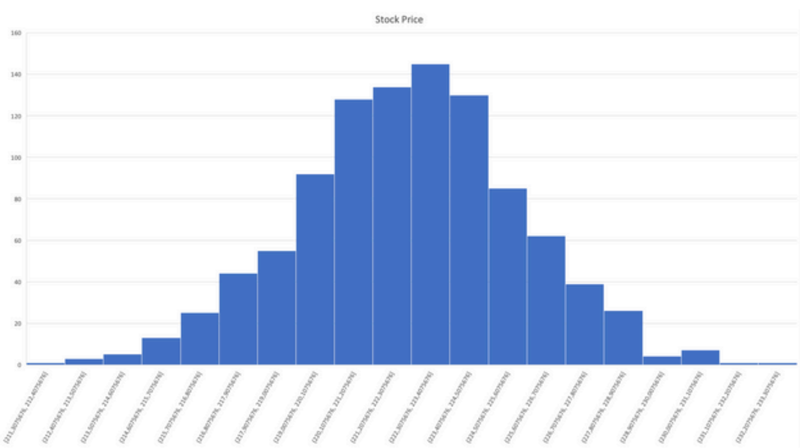
## 2.5 Provide a Monte-Carlo Simulation for question 3 assuming that the true expected growth rate of free cash flows in every year can differ from your baseline expectations by an error that is normally distributed with mean zero and standard deviation 2%. Please provide a histogram to summarize your new answer.

We extended our valuation work by running a Monte-Carlo simulation designed to reflect the natural uncertainty surrounding long-term growth assumptions. Instead of relying solely on a single deterministic trajectory, we introduced controlled randomness into the projected free cash flow growth rates to evaluate how small forecasting errors can influence the resulting equity value.
The simulation generated 1,000 alternative scenarios for the forecasted years, drawing growth deviations from a normal distribution with mean 0% and a standard deviation of 2%. From year 11 onward, the growth rate was kept constant to ensure coherence with the underlying model. Each simulated path produced a distinct set of free cash flows and a corresponding terminal value, which were automatically incorporated into the Excel model to compute the implied stock price for every iteration.

The distribution of simulated stock prices shows a bell-shaped pattern. Since Monte-Carlo outputs fluctuate at every run, we fixed one representative simulation to visualize the behaviour of the model under a specific set of random draws and to provide a consistent snapshot of how the valuation responds to growth-rate uncertainty. Values span approximately from \$211.3 to \$233.3, with the majority clustering in the \$222.3–\$223.4 range, indicating moderate dispersion around the central tendency.

The average outcome of \$222.22 lies close to the deterministic figure of \$226.05, while the interquartile range from \$220.21 to \$224.25 highlights a relatively tight concentration of results. A standard deviation of \$7.67 further suggests that the firm's valuation is only moderately affected by small variations in growth expectations.
Even under the most optimistic stock price around \$233, the simulated equity value remains significantly below the \$260.41 pre-leverage price. This outcome strengthens the conclusion that the chosen leverage structure does not generate shareholder value. Although the 20% debt ratio introduces tax benefits, these are insufficient to counterbalance the heightened distress and bankruptcy risk associated with higher leverage. Overall, the Monte-Carlo simulation confirms that, under the examined conditions, this capital structure does not optimize value for Salesforce.




| STOCK PRICE - MONTECARLO RESULTS | |
|---|---|
| Mean | 222,2236801 |
| Median | 222,3071419 |
| **1st Quartile (Q1)** | 220,2149152 |
| 3rd Quartile (Q3) | 224,2528658 |
| Standard Deviation | 7,668316641 |
| 5th Percentile | 216,8949543 |
| **95th Percentile** | 227,3578011 |

| For Comparison | |
|---|---|
| Value per stock (Pre Debt issued) | 260,40966 |
| **Value per stock (in case of Buyback)** | 226,05202 |

Figure 3: Results Montecarlo Simluation

# APPENDIX

## SOURCES

- 10K of Salesforce: https://investor.salesforce.com/financials/annual-reports/default.aspx
- Predictions adjustments: Capital IQ - CEDIF
- Rf: https://www.bloomberg.com/markets/rates-bonds
- Market Cap: https://finance.yahoo.com/quote/CRM/key-statistics/
- To compare Beta ((Software (System & Application)): https://pages.stern.nyu.edu/~adamodar/
- Risk Premium: https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histimpl.html

| TABLE 1 | Fraction of Unlevered firm value lost in bankruptcy |
|---|---|
| Food | 0.389 |
| Mining and Minerals | 0.463 |
| Oil | 0.364 |
| Clothing | 0.452 |
| Cons Durable | 0.422 |
| Chemicals | 0.435 |
| Drugs, Perfume, Tobacco | 0.532 |
| Construction | 0.374 |
| Steel | 0.369 |
| Fabricated Products | 0.35 |
| Machinery | 0.489 |
| Automobiles | 0.397 |
| Transportation | 0.413 |
| Retail Stores | 0.442 |
| Other | 0.474 |

Appendix 1: Fraction of Unlevered Firm Value Lost in Bankruptcy

| In millions | |
|---|---|
| Market Cap (July 2025) | $ 247.910,00 |
| Number of stocks (**N**) | 952 |
| Value per stocks | **$ 260,41** |
| | |
| Value of the Levered Firm (**VL**) | $ 215.201,52 |
| Debt Issued (initial debt=0) (**D**) | $ 42.983,41 |
| | |
| **New Stock Price (S)=** | VL/N |
| | **$ 226,05** |
| | |
| **Number of stocks acquired (X)=** | D/New Price |
| | **$ 190,15** |
| | |
| **Gain/loss per stock** | **$ -34,36** |
| **% gain/loss per stock** | **-13,19%** |

Appendix 2: Buyback operation